

TMFVC Assignment 13

Guangxin Zhao st194136

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Exercise 13.1

(a)

Code:

```
1 # Exercise 13.1
2 library(ggplot2)
3 library(dplyr)
4 library(car)
5
6 # Question (a)
7 # Load the data
8 data <- read.csv("vis_type.csv")
9
10 # Inspect the data
11 print(summary(data))
12
13 # Convert variables to factors
14 data$ID <- as.factor(data$ID)
15 data$visualization_type <- as.factor(data$visualization_type)
16 data$color <- as.factor(data$color)
17 data$reaction_time <- as.numeric(data$reaction_time)
```

Output:

	ID	visualization_type	color	reaction_time	
1	Min. :	1.0	Length:9888	Length:9888	Min. : 13.5
2	1st Qu.:	62.0	Class :character	Class :character	1st Qu.: 1033.1
3	Median :	125.5	Mode :character	Mode :character	Median : 1583.0
4	Mean :	125.0			Mean : 2250.1
5	3rd Qu.:	188.0			3rd Qu.: 2549.6
6	Max. :	246.0			Max. : 160128.8

(b)

Code:

```
1 # Question (b)
2 print(levels(data$visualization_type))
3
4 summary_visualization_type <- data %>%
5 group_by(visualization_type) %>%
6 summarise(mean_reaction_time = mean(reaction_time),
7           min_reaction_time = min(reaction_time),
8           max_reaction_time = max(reaction_time),
9           sd_reaction_time = sd(reaction_time))
10
11 print(summary_visualization_type)
12
13 # Remove extreme outliers
14 filtered_data <- data %>%
15 filter(reaction_time < quantile(reaction_time, 0.95))
16
17 # Boxplot
18 ggplot(filtered_data, aes(x = visualization_type, y = reaction_time, fill =
19   visualization_type)) +
20 geom_boxplot() +
21 labs(title = "Boxplot of Reaction Time (Without Extreme Outliers)",
22       x = "Visualization Type",
23       y = "Reaction Time") +
24 theme_minimal()
```

Output:

```
1 # A tibble: 3 × 5
2 visualization_type mean_reaction_time min_reaction_time max_reaction_time
3 <fct>                <dbl>                <dbl>                <dbl>
4 1 bar chart          2261.                83.4                160129.
5 2 line chart         2381.                13.5                75572.
6 3 pie chart          2108.                250                122128
7 # 1 more variable: sd_reaction_time <dbl>
```

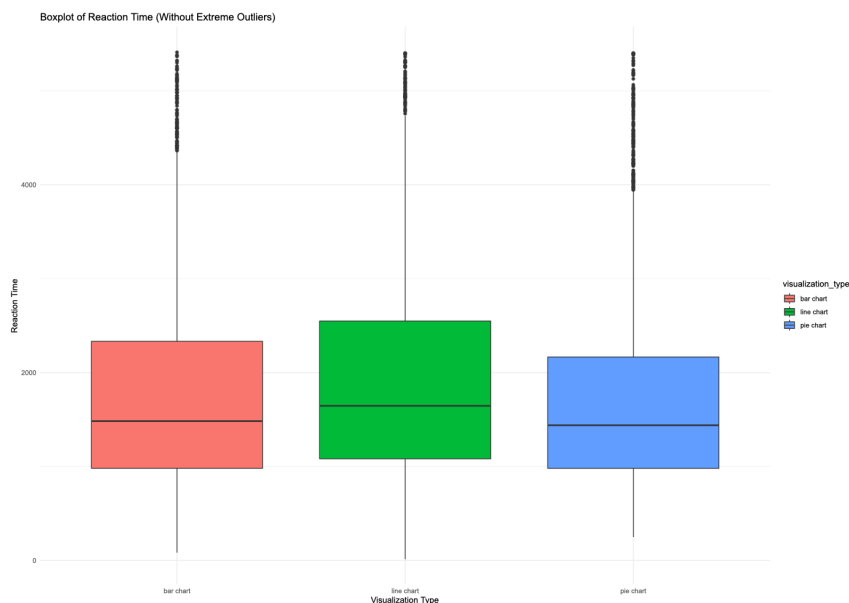


Figure 1: Visualization type vs. Reaction time

From both the boxplot (Fig 1) and the summary statistics, **pie charts** have the lowest average reaction time, while **line charts** show the highest average reaction time.

(c)

Code:

```
1 # Question (c)
2 print(levels(data$color))
3
4 summary_color <- data %>%
5 group_by(color) %>%
6 summarise(mean_reaction_time = mean(reaction_time),
7           min_reaction_time = min(reaction_time),
8           max_reaction_time = max(reaction_time),
9           sd_reaction_time = sd(reaction_time))
10
11 print(summary_color)
12
13 # Boxplot
14 ggplot(filtered_data, aes(x = color, y = reaction_time, fill = color)) +
15 geom_boxplot() +
16 labs(title = "Boxplot of Reaction Time by Color",
17       x = "Color", y = "Reaction Time") +
18 theme_minimal()
```

Output:

```
1 # A tibble: 4 × 5
2 color mean_reaction_time min_reaction_time max_reaction_time
3   <fct>          <dbl>          <dbl>          <dbl>
4   <dbl>
5 1 blue                2220.             208.          157142
6   ↳ 4318.
7 2 green                2223.             275.           75572.
8   ↳ 2852.
9 3 red                 2273.             13.5          160129.
10  ↳ 4182.
11 4 yellow              2284.             83.4          122128
12  ↳ 3496.
```

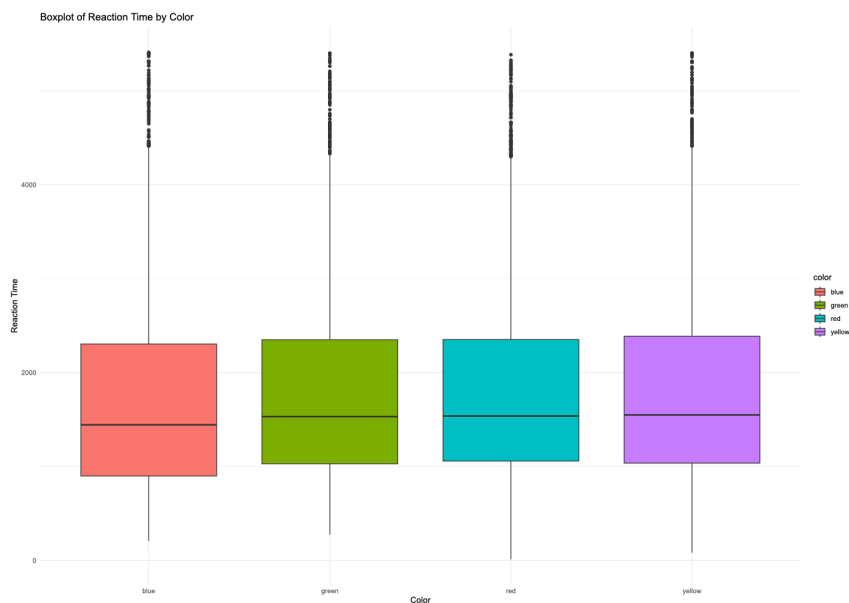


Figure 2: Color vs. Reaction time

From both the boxplot (Fig 2) and the summary statistics, **blue** yields the lowest average RT, while **yellow** yields the highest average RT.

(d)

Code:

```
1 # Question (d)
2 summary_data <- data %>%
3   group_by(visualization_type, color) %>%
4   summarise(mean_reaction_time = mean(reaction_time),
5             sd_reaction_time = sd(reaction_time),
6             n = n()) %>%
7   mutate(se = sd_reaction_time / sqrt(n),
8          lower_ci = mean_reaction_time - qt(0.975, n - 1) * se,
9          upper_ci = mean_reaction_time + qt(0.975, n - 1) * se)
10
11 print(summary_data)
12
13 # 95% CI plot
14 ggplot(summary_data, aes(x = visualization_type, y = mean_reaction_time, color
15   ↪ = color)) +
16   geom_point(size = 4, position = position_dodge(width = 0.5)) +
17   geom_errorbar(aes(ymin = lower_ci, ymax = upper_ci,
18     ↪ width = 0.2, position = position_dodge(width = 0.5)) +
19   labs(title = "95% Confidence Interval of Reaction Time",
20     ↪ x = "Visualization Type",
21     ↪ y = "Mean Reaction Time") +
22   theme_minimal()
```

Output:

```
1 # A tibble: 12 × 8
2 # Groups:   visualization_type [3]
3 visualization_type color mean_reaction_time sd_reaction_time n se
4 ↪ lower_ci
5 <fct> <fct> <dbl> <dbl> <int> <dbl>
6 ↪ <dbl>
7 1 bar chart blue 2437. 5985. 824 209.
8 ↪ 2028.
9 2 bar chart green 2264. 2882. 824 100.
10 ↪ 2067.
11 3 bar chart red 2074. 5766. 824 201.
12 ↪ 1680.
13 4 bar chart yellow 2268. 2455. 824 85.5
14 ↪ 2100.
15 5 line chart blue 2257. 2415. 824 84.1
16 ↪ 2092.
17 6 line chart green 2480. 3538. 824 123.
18 ↪ 2238.
19 7 line chart red 2420. 3574. 824 124.
20 ↪ 2175.
21 8 line chart yellow 2369. 2220. 824 77.3
22 ↪ 2217.
23 9 pie chart blue 1966. 3769. 824 131.
24 ↪ 1708.
25 10 pie chart green 1926. 1854. 824 64.6
26 ↪ 1799.
27 11 pie chart red 2326. 2533. 824 88.2
28 ↪ 2153.
29 12 pie chart yellow 2215. 5071. 824 177.
30 ↪ 1868.
31 # 1 more variable: upper_ci <dbl>
```

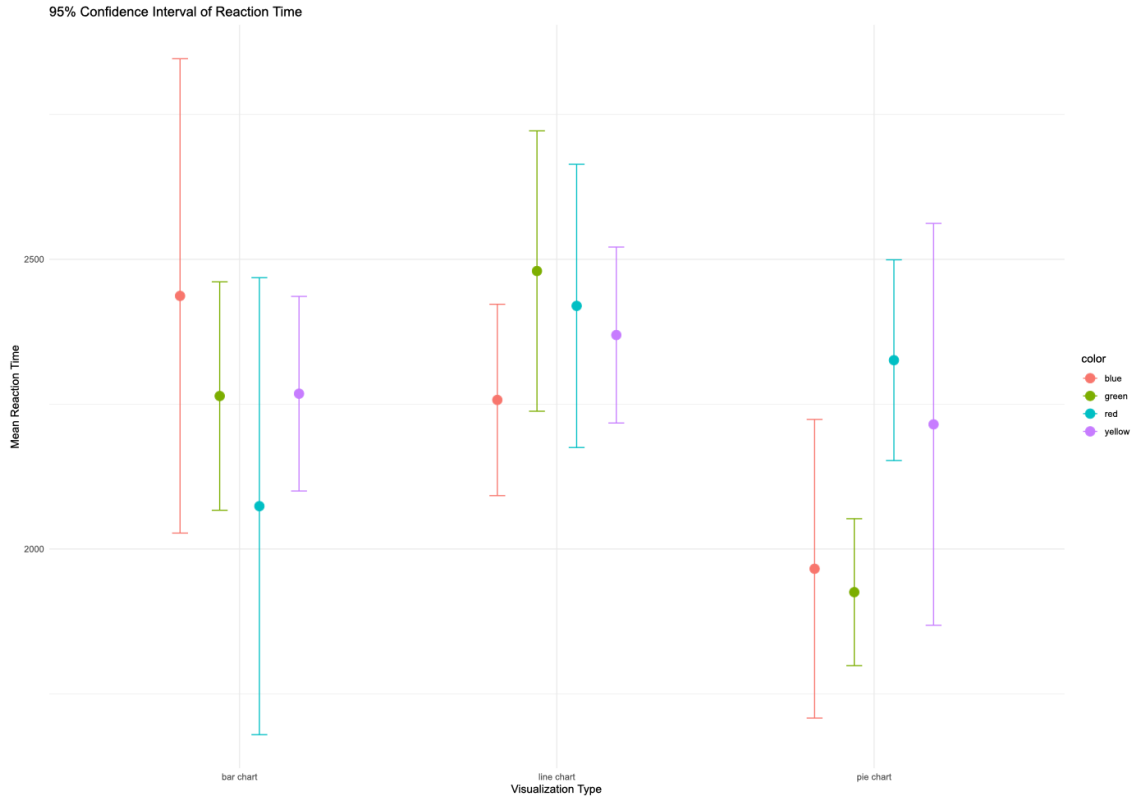


Figure 3: Visualization type and color vs. Reaction time

Fig 3 shows the 95% confidence interval of the mean reaction time for each combination of visualization type and color.

(e)

Based on the summary statistics and Fig 3, we can conclude that the visualization type and color have a ~~significant~~ impact on the reaction time. The **pie chart** with **green** color has the lowest average reaction time, while the **line chart** with **green** color has the highest average reaction time.

Exercise 13.2

(a) 

- **Independence of observations:** The data in each group should be independent of data in other groups.
- **Normality:** ~~The residuals~~ of the data within each group should be normally distributed.
- **Homogeneity of variances:** The variance within each group should be the same across all groups.

Code:

```
1 # Exercise 13.2
2 library(nortest)
3 library(ARTool)
4 library(emmeans)
5
6 # Question (a)
7 # Check the independence of the data
8 chisq.test(table(data$visualization_type, data$color))
9
10 # Check the normality of the data
11 ad.test(data$reaction_time)
12
13 # Check the homogeneity of variance
14 leveneTest(reaction_time ~ visualization_type * color, data = data)
```

Output:

```
1 Pearson's Chi-squared test
2
3 data:  table(data$visualization_type, data$color)
4 X-squared = 0, df = 6, p-value = 1
5
6
7      Anderson-Darling normality test
8
9 data:  data$reaction_time
10 A = 1549.2, p-value < 2.2e-16
11
12 Levene's Test for Homogeneity of Variance (center = median)
13   Df F value Pr(>F)
14 group  11  1.1474 0.3189
15      9876
```

(b) 

- ~~**Independence of observations:** The p-value of the chi-squared test is 1, which indicates that the data is independent.~~
- **Normality:** The p-value of the Anderson-Darling normality test is less than 0.05, which indicates that the data is **not** normally distributed.
- **Homogeneity of variances:** The p-value of the Levene's test is 0.3189, which indicates that the variances are homogeneous.

In order to make the data meet the normality assumption, we can apply a transformation to the data. For example, we can apply **log transformation** and **square root transformation** to the reaction time data. Alternatively, we can **remove the extreme outliers** from the data.

Code:

```
1 # Question (b)
2 # Log transformation
3 data$log_reaction_time <- log(data$reaction_time)
4
5 # Square root transformation
6 data$sqrt_reaction_time <- sqrt(data$reaction_time)
7
8 # Remove extreme outliers
9 Q1 <- quantile(data$reaction_time, 0.25)
10 Q3 <- quantile(data$reaction_time, 0.75)
11 IQR <- Q3 - Q1
12 lower_bound <- Q1 - 1.5 * IQR
13 upper_bound <- Q3 + 1.5 * IQR
14 data_filtered <- data[data$reaction_time >= lower_bound & data$reaction_time
  ↪ <= upper_bound, ]
```

(c)



Code:

```
1 # Question (c)
2 # Log transformation
3 ad.test(data$log_reaction_time)
4 # Square root transformation
5 ad.test(data$sqrt_reaction_time)
6 # Remove extreme outliers
7 ad.test(data_filtered$reaction_time)
8
9 # Generate the Q-Q plots
10 par(mfrow = c(2, 2))
11 qqPlot(data$reaction_time, distribution = "norm",
12        main = "Q-Q Plot of Reaction Time")
13 qqPlot(data$log_reaction_time, distribution = "norm",
14        main = "Q-Q Plot of Log Reaction Time")
15 qqPlot(data$sqrt_reaction_time, distribution = "norm",
16        main = "Q-Q Plot of Square Root Reaction Time")
17 qqPlot(data_filtered$reaction_time, distribution = "norm",
18        main = "Q-Q Plot (Without Extreme Outliers)")
19 par(mfrow = c(1,1))
```

Output:

```
1      Anderson-Darling normality test
2
3 data:  data$log_reaction_time
4 A = 19.208, p-value < 2.2e-16
5
6
7      Anderson-Darling normality test
8
9 data:  data$sqrt_reaction_time
10 A = 330.28, p-value < 2.2e-16
11
12
13      Anderson-Darling normality test
14
15 data:  data_filtered$reaction_time
16 A = 221.45, p-value < 2.2e-16
```

Even after trying the **log transform**, the **square-root transform**, and **removing extreme outliers**, the Anderson-Darling tests all return highly significant p-values, which indicates that the data are still not coming from a normal distribution. So in a strict sense, the **normality assumption** for ANOVA is still **violated**.

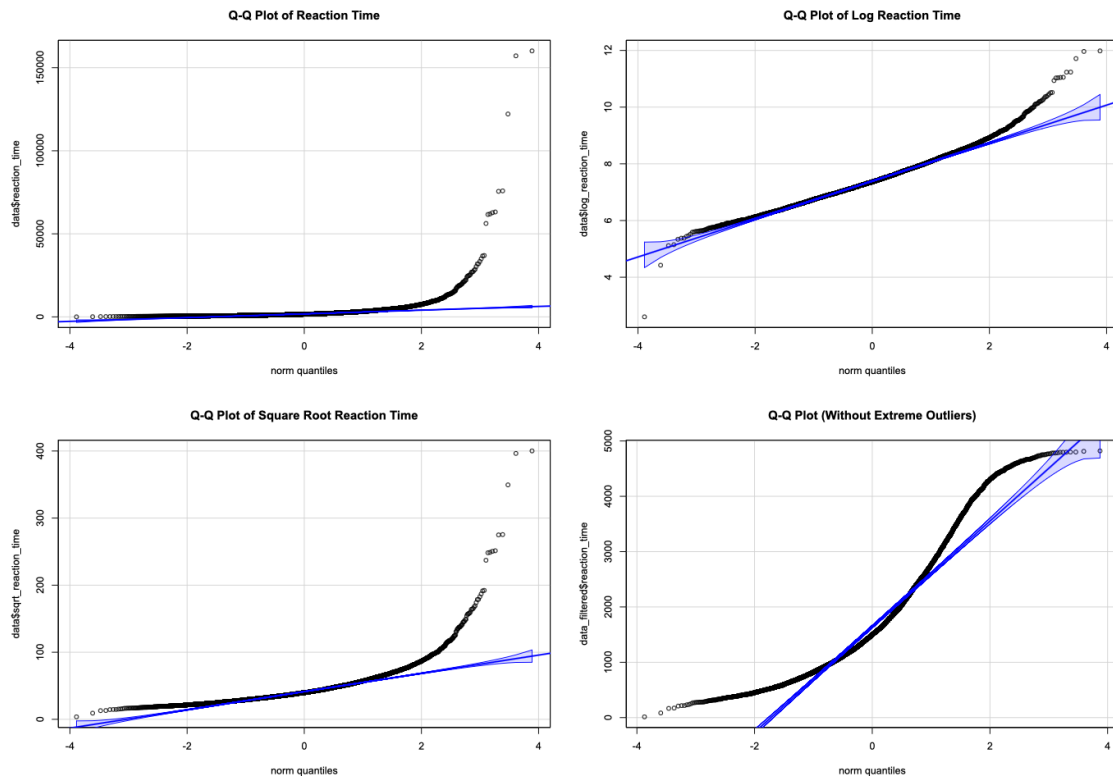


Figure 4: Q-Q plots of different transformations

However, in practice, with very large samples, ANOVA may still be relatively robust to non-normality. According to Fig 4, the **log transformation** shows a better approximation to normality, even though some deviations exist in the tails. So we can try to perform ANOVA on the **log transformed** data. Meanwhile, we can also perform Aligned Rank Transform ANOVA (ART ANOVA), which is useful when the data is not normally distributed.

(d) 

Let's first perform ANOVA on the log-transformed data:

Code:

```
1 # Question (d)
2 # ANOVA on the log-transformed data
3 anova_log <- aov(log_reaction_time ~ visualization_type * color, data = data)
4 summary(anova_log)
```

Output:

```
1 Df Sum Sq Mean Sq F value Pr(>F)
2 visualization_type      2      33 16.706 33.987 1.95e-15 ***
3 color                   3      14  4.645  9.449 3.13e-06 ***
4 visualization_type:color  6      24  4.079  8.299 5.45e-09 ***
5 Residuals              9876    4854   0.492
6 ---
7 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The results of ANOVA on the log-transformed data show that both the **visualization type** and **color** have a significant effect on reaction time. Additionally, there is a significant interaction effect between **visualization type** and **color** on reaction time.

Let's then perform ART ANOVA on the original data:

Code:

```
1 # ART ANOVA on the original data
2 model_art <- art(reaction_time ~ visualization_type * color, data = data)
3 anova(model_art)
```

Output:

```
1 Analysis of Variance of Aligned Rank Transformed Data
2
3 Table Type: Anova Table (Type III tests)
4 Model: No Repeated Measures (lm)
5 Response: art(reaction_time)
6
7           Df Df.res F value    Pr(>F)
8 1 visualization_type      2   9876 38.9192 < 2.22e-16 ***
9 2 color                  3   9876  8.4038 1.4154e-05 ***
10 3 visualization_type:color  6   9876  9.1834 4.7260e-10 ***
11 ---
12 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The results of ART ANOVA on the original data align with the results of ANOVA on the log-transformed data. Both the **visualization type** and **color** have a significant effect on reaction time, and there is a significant interaction effect between **visualization type** and **color** on reaction time.

(e)

Code 

```
1 # Question (e)
2 # Post-hoc test
3 posthoc_interaction <- emmeans(
4   artlm(model_art, "visualization_type:color"),
5   ~ visualization_type:color,
6   adjust = "bonferroni"
7 )
8
9 df_means <- as.data.frame(posthoc_interaction)
10
11 print(df_means)
```

Output:

```
1 visualization_type color    emmean      SE    df lower.CL upper.CL
2 bar chart          blue  4839.808 99.06199 9876 4555.904 5123.713
3 line chart         blue  4740.322 99.06199 9876 4456.417 5024.226
4 pie chart          blue  4753.953 99.06199 9876 4470.048 5037.857
5 bar chart          green  5061.949 99.06199 9876 4778.044 5345.854
6 line chart         green  5119.353 99.06199 9876 4835.449 5403.258
7 pie chart          green  5061.835 99.06199 9876 4777.930 5345.740
8 bar chart          red    4390.954 99.06199 9876 4107.049 4674.858
9 line chart         red    4856.504 99.06199 9876 4572.599 5140.408
10 pie chart         red    5565.591 99.06199 9876 5281.686 5849.496
11 bar chart         yellow 4972.508 99.06199 9876 4688.604 5256.413
12 line chart        yellow 4933.414 99.06199 9876 4649.509 5217.318
13 pie chart        yellow 5037.809 99.06199 9876 4753.905 5321.714
14
15 Confidence level used: 0.95
16 Conf-level adjustment: bonferroni method for 12 estimates
```



Code:

```
1 # 95% CI plot
2 ggplot(df_means, aes(x = visualization_type, y = emmean, color = color)) +
3   geom_point(position = position_dodge(width = 0.5), size = 2) +
4   geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL),
5                 position = position_dodge(width = 0.5), width = 0.2) +
6   labs(x = "Visualization Type",
7        y = "Aligned Rank Means (from ART)",
8        title = "95% CIs by Visualization Type and Color") +
9   theme_bw()
```

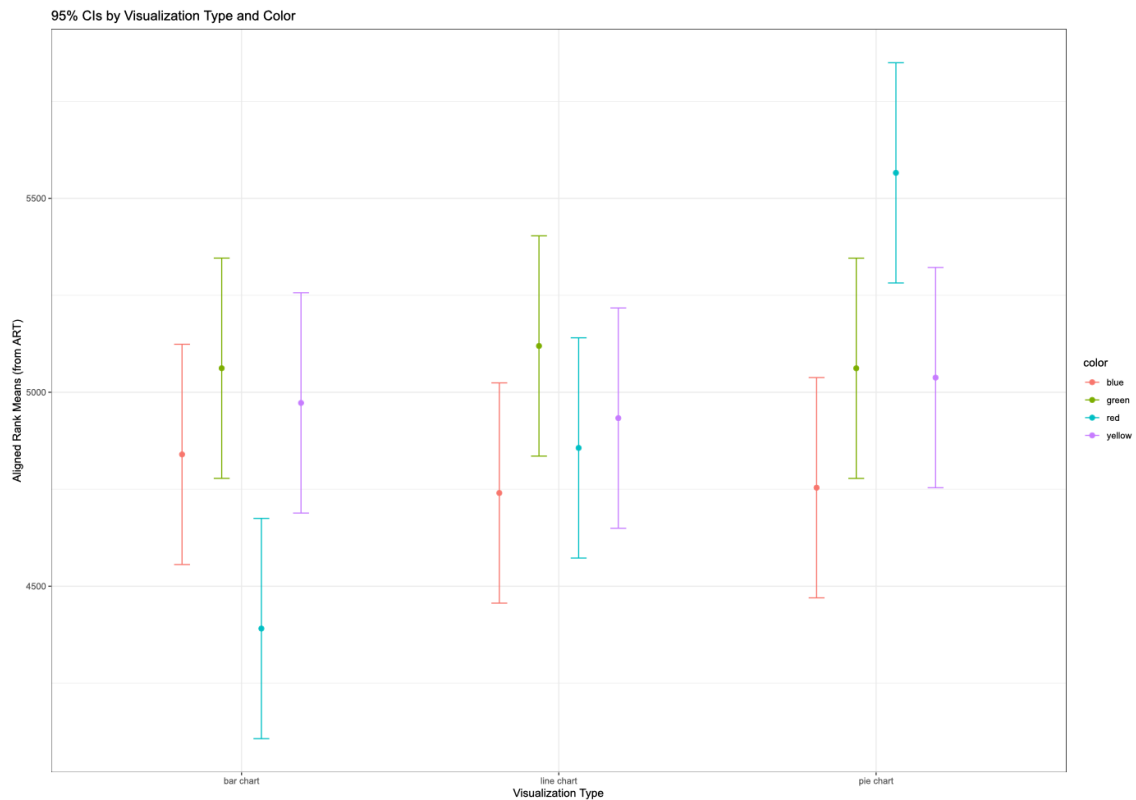


Figure 5: 95% CIs by Visualization Type and Color

Based on the summary statistics and Fig 5, we can conclude that the visualization type and color have a significant impact on the reaction time. The **bar chart** with **red** color has the lowest average reaction time, while the **pie chart** with red color has the highest average reaction time.